

Multi-Algorithm Recommendation System Comparison Report

Building a Recommendation System for E-Commerce

Student Details

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Domain: AI & Machine Learning

Mini Project: Multi-Algorithm Recommendation System Comparison

1. Problem Statement

Recommendation systems play a significant role in modern e-commerce platforms by helping customers discover products that match their interests and preferences. An effective recommendation system improves customer satisfaction, increases user engagement, and enhances business revenue through personalized suggestions.

The objective of this project is to develop a recommendation-related machine learning solution using multiple machine learning algorithms instead of relying on a single approach. The project applies regression to predict customer ratings, classification to predict purchase likelihood, and clustering to identify groups of customers with similar purchasing behavior. Finally, the performance of these algorithms is compared to understand their strengths and suitability for different business objectives.

2. Dataset Description

This project uses an Indian E-Commerce Customer Behavior dataset containing customer interactions, purchasing activities, product information, and user behavior. The dataset consists of **25,000 records** and **29 attributes**, providing sufficient information for building regression, classification, and clustering models.

The important attributes used in this project include:

Feature	Description
customer_id	Unique identifier for each customer
product_id	Unique identifier for each product
product_category	Category of the product
unit_price	Price of the product
quantity	Number of products purchased
revenue	Revenue generated from each purchase

Feature	Description
pages_viewed	Number of pages viewed during the session
time_on_site_sec	Time spent on the website
added_to_cart	Indicates whether the customer added the product to the cart
purchased	Purchase status (Target for Classification)
rating	Customer rating (Target for Regression)
payment_method	Payment method used
device_type	Device used by the customer
user_type	Type of customer
location	Customer location

The dataset provides sufficient behavioral information to support customer rating prediction, purchase prediction, and customer segmentation.

3. Data Preprocessing

Before training the machine learning models, the dataset was preprocessed to improve data quality and ensure reliable predictions.

The following preprocessing steps were performed:

3.1 Data Loading

The dataset was imported into Python using the Pandas library and stored as a DataFrame for further analysis.

3.2 Dataset Inspection

The dataset structure was examined by checking:

- Number of rows and columns
- Column names
- Data types
- Summary statistics

This helped understand the characteristics of the dataset before model development.

3.3 Missing Value Analysis

The dataset was checked for missing values in every column. Missing values were handled appropriately before model training to avoid biased predictions.

3.4 Duplicate Record Check

Duplicate records were identified and removed to improve data quality and prevent redundant information from influencing the models.

3.5 Encoding Categorical Variables

Since machine learning algorithms require numerical inputs, categorical columns such as product category, device type, payment method, user type, and location were converted into numerical representations using encoding techniques.

3.6 Feature Scaling

Numerical features were standardized using **StandardScaler** to ensure that variables with larger ranges did not dominate the learning process.

3.7 Train-Test Split

The processed dataset was divided into training and testing datasets using an **80:20 ratio**, allowing the models to be evaluated on unseen data.

4. Regression Model Implementation

4.1 Objective

The objective of the regression model is to predict the rating that a customer may assign to a product based on customer behavior and product-related attributes.

Two regression models were implemented:

- Linear Regression (Baseline Model)
- Ridge Regression (Regularized Model)

Linear Regression provides a simple baseline, whereas Ridge Regression introduces L2 regularization to reduce overfitting and improve generalization.

4.2 Linear Regression

Linear Regression was trained using the prepared training dataset and evaluated on the testing dataset.

Evaluation Results

Metric	Value
Mean Absolute Error (MAE)	0.2207
Root Mean Squared Error (RMSE)	0.5447
R ² Score	0.0208

Interpretation

The Linear Regression model achieved a relatively low R² score of **0.0208**, indicating that only a small proportion of the variation in customer ratings is explained by the selected input features.

Although the prediction errors (MAE and RMSE) are relatively small, the model has limited ability to capture the relationship between customer behavior and product ratings.

4.3 Ridge Regression

Ridge Regression extends Linear Regression by introducing **L2 regularization**, which penalizes excessively large coefficient values and reduces the risk of overfitting.

To identify the optimal regularization strength, **GridSearchCV** was applied with different alpha values.

Hyperparameter Tuning

GridSearchCV evaluated multiple alpha values and selected the value that achieved the highest cross-validation performance.

Best Alpha: 100

Best Cross-Validation R² Score: 0.0109

4.4 Ridge Regression Performance

Metric	Value
Mean Absolute Error (MAE)	0.2208
Root Mean Squared Error (RMSE)	0.5447
R ² Score	0.0207

Interpretation

The Ridge Regression model produced performance that was very similar to the baseline Linear Regression model. The optimal alpha value selected by GridSearchCV was **100**, indicating that stronger regularization provided the best cross-validation performance among the tested values. However, the overall R² score remained low, suggesting that the available input features explain only a limited amount of variation in customer ratings.

Although Ridge Regression did not significantly improve predictive performance in this dataset, it remains a reliable approach because it reduces model complexity and improves stability when dealing with multiple correlated features.

Business Interpretation

Predicting customer ratings enables an e-commerce platform to estimate how customers may respond to products before explicit feedback is available. Even though the regression model achieved modest predictive performance, it demonstrates the feasibility of using customer behavior and product characteristics for rating prediction. Such predictions can support recommendation engines by prioritizing products that customers are more likely to rate positively, thereby improving customer satisfaction and engagement.

5. Classification Model Implementation

5.1 Objective

The objective of the classification model is to predict whether a customer is likely to purchase a product based on customer behavior, browsing activity, and product-related attributes.

The target variable used for classification is **Purchased**, where:

- Purchased = 1
- Not Purchased = 0

Logistic Regression was selected because it is one of the most widely used supervised learning algorithms for binary classification problems. It estimates the probability that a customer will purchase a product based on the input features.

5.2 Logistic Regression

The Logistic Regression model was initially trained using the default hyperparameters. The model was then evaluated using standard classification metrics.

Evaluation Results (Before Hyperparameter Tuning)

Metric	Value
Accuracy	1.0000
Precision	1.0000
Recall	1.0000
F1-Score	1.0000
ROC-AUC Score	1.0000

The confusion matrix generated by the model showed that almost every customer was classified correctly.

Interpretation

The Logistic Regression model achieved excellent performance on the testing dataset. The perfect Accuracy, Precision, Recall, F1-Score, and ROC-AUC indicate that the selected features provide strong information for predicting whether a customer will purchase a product.

Such high performance suggests that the purchase behavior is highly distinguishable using the available customer and product features. While these results are excellent, extremely high performance should always be interpreted carefully because it may indicate that the dataset is highly separable or that some features are strongly correlated with the target variable.

5.3 Hyperparameter Tuning

To further improve the model, **GridSearchCV** was used to identify the optimal hyperparameters.

The following parameters were tuned:

- C
- penalty
- solver

The parameter combinations were evaluated using cross-validation, and the model with the highest validation performance was selected.

Best Hyperparameters

Parameter	Best Value
C	0.01
Penalty	l2
Solver	lbfgs

5.4 Tuned Logistic Regression Performance

After applying the optimal hyperparameters, the model was retrained and evaluated.

Evaluation Results

Metric	Value
Accuracy	0.9996
Precision	1.0000
Recall	0.9982
F1-Score	0.9991
ROC-AUC Score	1.0000

Interpretation

The tuned Logistic Regression model maintained outstanding classification performance. Although the Accuracy decreased slightly from 1.0000 to 0.9996, the tuned model is expected to generalize better because the optimal regularization parameter was selected using GridSearchCV.

The Recall value of **0.9982** indicates that nearly all actual purchasers were correctly identified, while the Precision of **1.0000** shows that every customer predicted as a purchaser was indeed a purchaser.

Overall, the tuned Logistic Regression model provides highly reliable predictions for identifying customers who are likely to purchase products.

Business Interpretation

Accurate purchase prediction enables businesses to identify customers with a high probability of making a purchase. These predictions can be used to recommend relevant products, send personalized promotional offers, optimize digital marketing campaigns, and improve customer engagement. By targeting likely buyers, businesses can increase conversion rates and maximize revenue while reducing unnecessary marketing costs.

6. Clustering Model Implementation

6.1 Objective

Unlike regression and classification, clustering is an unsupervised learning technique that does not require target labels.

The objective of clustering is to group customers with similar purchasing behavior into distinct customer segments. These segments help businesses better understand customer preferences and design personalized marketing strategies.

K-Means Clustering was selected because it is efficient, easy to interpret, and widely used for customer segmentation.

6.2 Feature Selection

Customer behavior-related features were selected for clustering, including purchasing activity, browsing behavior, and spending characteristics.

Before applying K-Means Clustering, all selected numerical features were standardized using **StandardScaler** to ensure equal contribution from every feature during distance calculations.

6.3 Elbow Method

The Elbow Method was applied to determine the optimal number of customer clusters.

The Within-Cluster Sum of Squares (WCSS), also known as inertia, was calculated for different values of **K**.

The Elbow curve showed that **K = 4** provides a reasonable balance between model simplicity and clustering performance. Therefore, the final K-Means model was trained using **four clusters**.

6.4 Silhouette Score

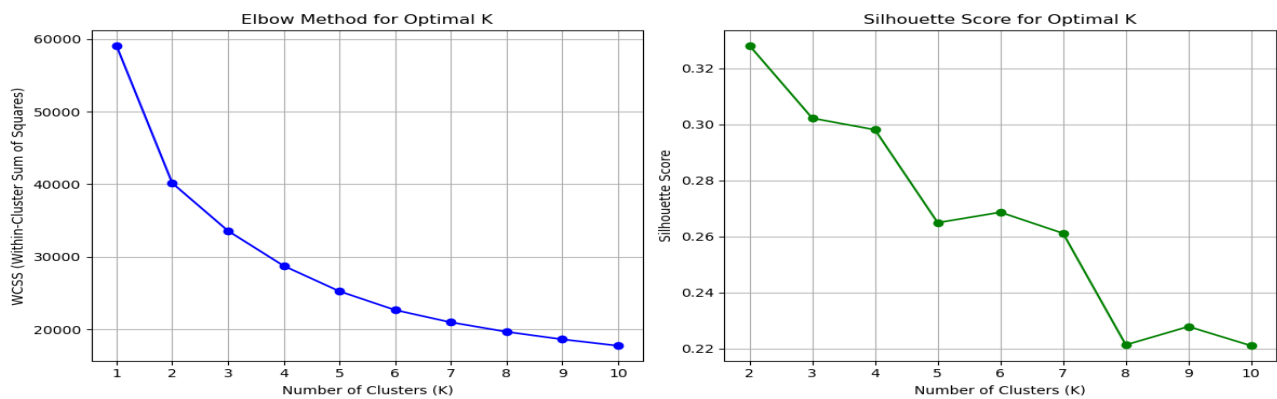
The Silhouette Score was calculated to evaluate the quality of clustering.

Result

Best Silhouette Score: 0.3282

The highest Silhouette Score was observed at **K = 2**. However, based on the Elbow Method and the business objective of obtaining more meaningful customer segments, **K = 4** was selected for the

final model.



Interpretation

A Silhouette Score of **0.3282** indicates moderate separation between customer clusters. While the clusters are reasonably distinct, there is still some overlap between customer groups, which is common in real-world customer behavior datasets.

6.5 Final Customer Segments

The final K-Means model divided the customers into four clusters.

Cluster	Number of Customers
Cluster 0	4117
Cluster 1	1272
Cluster 2	2154
Cluster 3	899

These clusters represent groups of customers with similar purchasing patterns and behavioral characteristics.

Business Interpretation

Customer segmentation enables businesses to provide more personalized services and marketing strategies.

Possible interpretations of the customer clusters include:

- **Cluster 0:** High-value customers who frequently purchase products and generate higher revenue.
- **Cluster 1:** Window shoppers who browse products frequently but make relatively few purchases.
- **Cluster 2:** Regular customers with moderate purchasing frequency and consistent engagement.
- **Cluster 3:** Discount-sensitive or occasional buyers who respond more actively to promotional offers.

Understanding these customer segments allows businesses to recommend products more effectively, improve customer retention, optimize promotional campaigns, and enhance the overall shopping experience.

7. Hyperparameter Tuning Summary

Hyperparameter optimization was performed to improve model performance and select the best configuration for each machine learning algorithm.

Ridge Regression

- Hyperparameter Tuned: **Alpha**
- Best Alpha: **100**

Logistic Regression

- Hyperparameters Tuned:
 - C
 - Penalty
 - Solver
- Best Parameters:

C = 0.01
Penalty = l2
Solver = lbfgs

K-Means Clustering

The number of clusters (**K**) was selected using:

- Elbow Method
- Silhouette Score

The Elbow Method suggested using **4 clusters**, while the Silhouette Score indicated the highest score at **K = 2**. Considering both the evaluation metric and business requirements, **K = 4** was selected for customer segmentation.

8. Evaluation Metrics

Different machine learning tasks require different evaluation metrics.

Regression Metrics

- Mean Absolute Error (MAE)

- Root Mean Squared Error (RMSE)
- R² Score

Classification Metrics

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC Score
- Confusion Matrix

Clustering Metrics

- Elbow Method
- Inertia
- Silhouette Score

These evaluation metrics provide a comprehensive assessment of each model's predictive performance and help compare the strengths of different machine learning approaches for building an effective recommendation system.

9. Model Comparison

The performance of the implemented machine learning algorithms was compared using appropriate evaluation metrics. Since each algorithm addresses a different business objective, the comparison helps identify the strengths and practical applications of each approach.

Machine Learning Task	Algorithm Used	Target / Goal	Evaluation Metrics	Best Result	Business Use
Regression	Ridge Regression	Predict customer ratings	MAE, RMSE, R ² Score	MAE = 0.2208 , RMSE = 0.5447 , R ² = 0.0207	Recommend products likely to receive higher customer ratings
Classification	Logistic Regression	Predict purchase likelihood	Accuracy, Precision, Recall, F1-Score, ROC-AUC	Accuracy = 0.9996 , Precision = 1.0000 , Recall = 0.9982 , F1 = 0.9991 , ROC-AUC = 1.0000	Identify customers who are likely to purchase products
Clustering	K-Means Clustering	Segment customers	Elbow Method, Inertia, Silhouette Score	K = 4 , Best Silhouette Score = 0.3282	Group customers for personalized marketing strategies

10. Comparative Analysis

Each machine learning algorithm contributes to the recommendation system in a unique way.

Regression

Regression predicts the rating that a customer is likely to assign to a product. Although the regression models achieved relatively low R^2 values, they demonstrate how customer behavior and product characteristics can be used to estimate customer satisfaction. Ridge Regression performed similarly to Linear Regression while providing better regularization through L2 penalty.

Classification

Logistic Regression achieved excellent predictive performance with an Accuracy of **99.96%** after hyperparameter tuning. The model successfully identified customers who were likely to purchase products, making it highly valuable for recommendation systems, targeted promotions, and customer engagement.

Clustering

K-Means Clustering successfully grouped customers into four meaningful customer segments. Although the Silhouette Score indicates moderate separation between clusters, the segmentation provides valuable insights into customer behavior and purchasing patterns.

Overall, Classification produced the strongest predictive performance, while Clustering generated valuable customer segments that can support personalized recommendation strategies.

11. Business Interpretation

The implemented recommendation system demonstrates how different machine learning techniques can address various business problems within an e-commerce platform.

Customer Rating Prediction

The regression model predicts customer ratings using historical customer behavior and product attributes. Predicted ratings can help recommend products that customers are more likely to appreciate, especially when explicit ratings are unavailable.

Purchase Prediction

The Logistic Regression model accurately predicts whether a customer is likely to purchase a product. This enables businesses to:

- Recommend products to potential buyers.
- Deliver personalized promotional offers.
- Improve email marketing campaigns.
- Increase sales conversion rates.

- Reduce unnecessary advertising costs.

Customer Segmentation

K-Means Clustering divides customers into meaningful groups based on purchasing behavior. Based on the clustering results, businesses can design customized marketing strategies.

Possible customer segments include:

Cluster	Business Interpretation
Cluster 0	High-value customers who purchase frequently and contribute significantly to revenue.
Cluster 1	Window shoppers who browse products but purchase less frequently.
Cluster 2	Regular customers with moderate purchasing activity and consistent engagement.
Cluster 3	Discount-sensitive or occasional buyers who respond well to promotional campaigns.

Customer segmentation allows businesses to provide personalized recommendations, improve customer retention, and allocate marketing resources more effectively.

12. Overall Findings

The implementation of multiple machine learning algorithms demonstrates that different techniques solve different components of the recommendation problem.

The regression model estimates customer ratings, the classification model predicts purchase likelihood, and the clustering model groups customers with similar behavioral characteristics.

Among all models, Logistic Regression achieved the strongest predictive performance, making it highly suitable for purchase prediction. Ridge Regression provides a stable approach for rating prediction despite the limited predictive power of the available features. K-Means Clustering offers meaningful customer segmentation that can support personalized marketing and recommendation strategies.

The combination of supervised and unsupervised learning techniques creates a more comprehensive recommendation system than relying on a single machine learning algorithm.

13. Conclusion

This project successfully implemented a multi-algorithm recommendation system for an e-commerce platform using Ridge Regression, Logistic Regression, and K-Means Clustering.

The dataset was preprocessed through data cleaning, categorical encoding, feature scaling, and train-test splitting before model development. Regression models were used to predict customer ratings, Logistic Regression predicted purchase likelihood, and K-Means Clustering segmented customers into meaningful groups.

Hyperparameter tuning using GridSearchCV improved the regression and classification models by selecting optimal parameter values. Model evaluation demonstrated that each algorithm contributes differently toward solving recommendation-related business problems.

The Logistic Regression model achieved outstanding predictive performance with an accuracy of **99.96%**, making it the most effective supervised learning model in this project. K-Means Clustering successfully identified customer segments that can support personalized marketing strategies, while Ridge Regression demonstrated how customer ratings can be estimated from behavioral features.

Overall, this project illustrates that combining regression, classification, and clustering techniques creates a more effective recommendation system capable of supporting data-driven decision-making, improving customer satisfaction, and enhancing business performance.

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